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Disaster Alert & Community Response App

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Problem Statement

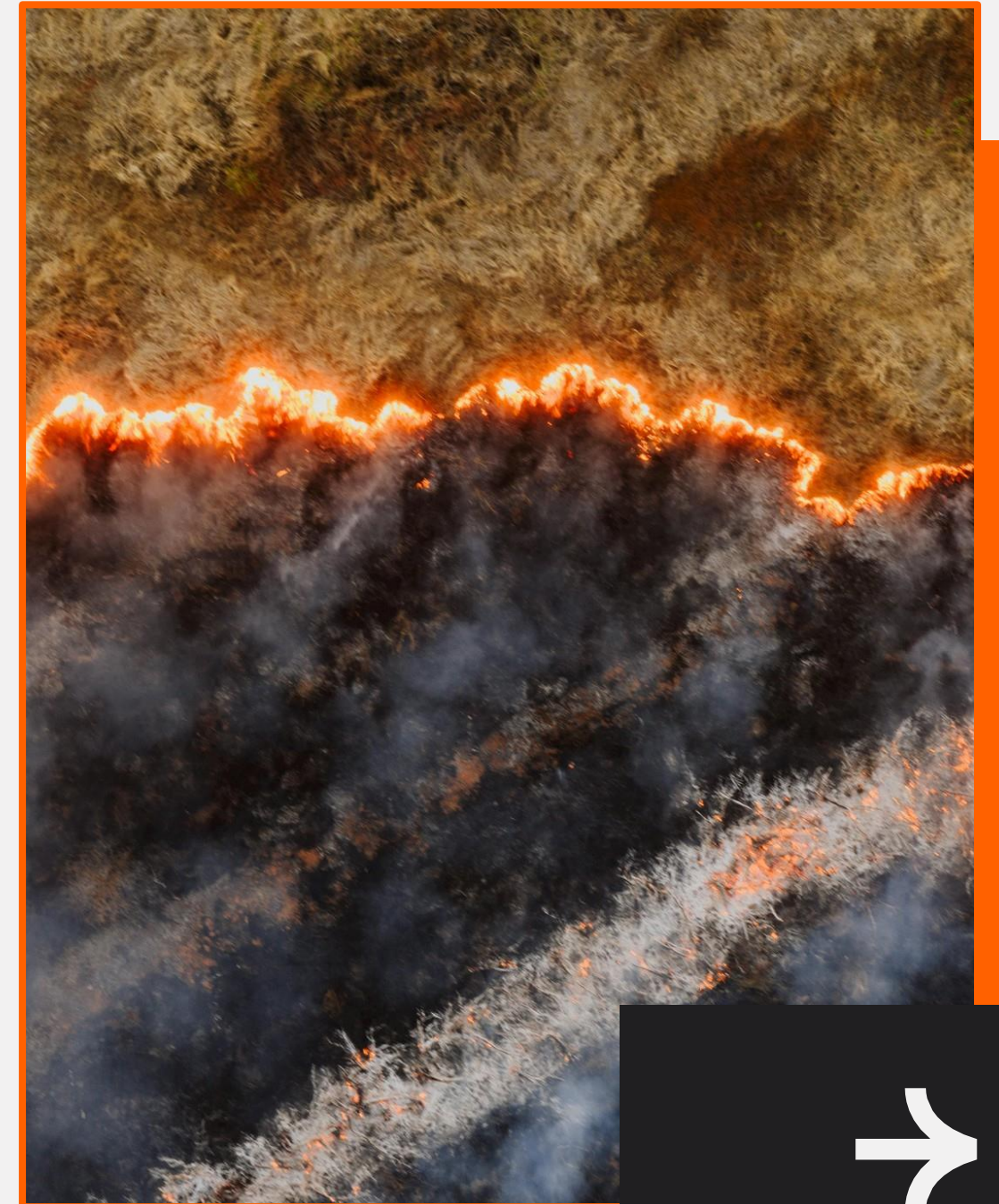
- During disasters, people may not receive timely alerts.
- Incident reporting and location sharing can be difficult.
- Affected people may struggle to find nearby assistance.
- Communication between citizens and response teams can be fragmented.
- A centralized platform is needed for faster information sharing and coordination.





Objectives

- Provide timely disaster alerts to users.
- Allow users to report emergency incidents with location.
- Enable users to request emergency assistance.
- Support volunteers and community members in helping affected people.
- Help users find nearby shelters and emergency services.
- Provide official announcements and updates during disasters.
- Enable map-based disaster tracking.
- Improve communication, coordination, and community response during emergencies.



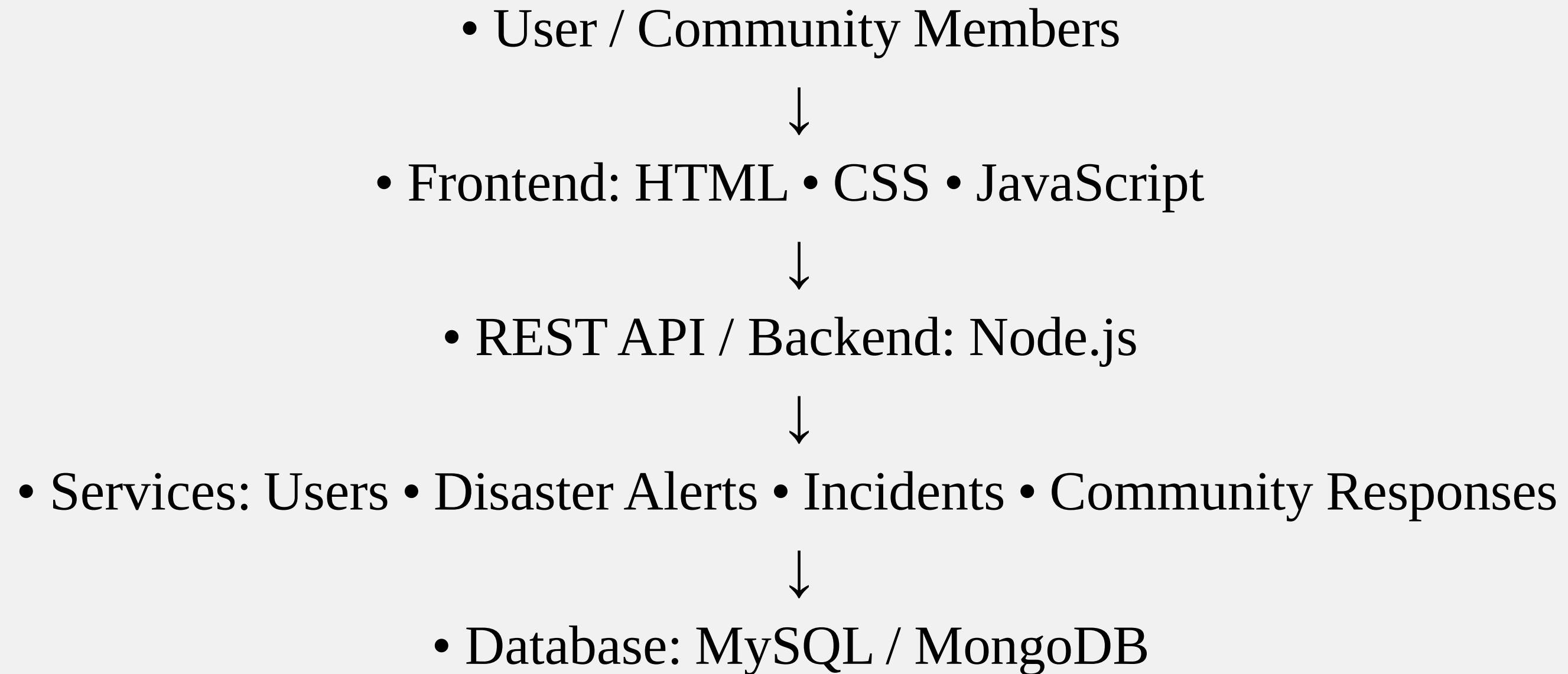


Proposed System

- Centralized disaster communication and response platform.
- Users can receive alerts, report incidents and request help.
- Community members can offer assistance to affected people.
- Map-based features support location-aware disaster tracking.
- Backend services manage users, alerts, incidents and responses.
- Authentication and authorization help protect user and emergency information.



System Architecture





Modules / Functionalities

1. User & Emergency Contact Management
2. Real-time Disaster Alerts
3. Incident Reporting with Location
4. Emergency Help Requests
5. Volunteer / Community Assistance
6. Nearby Shelters & Emergency Services
7. Official Announcements & Updates
8. Map-based Disaster Tracking



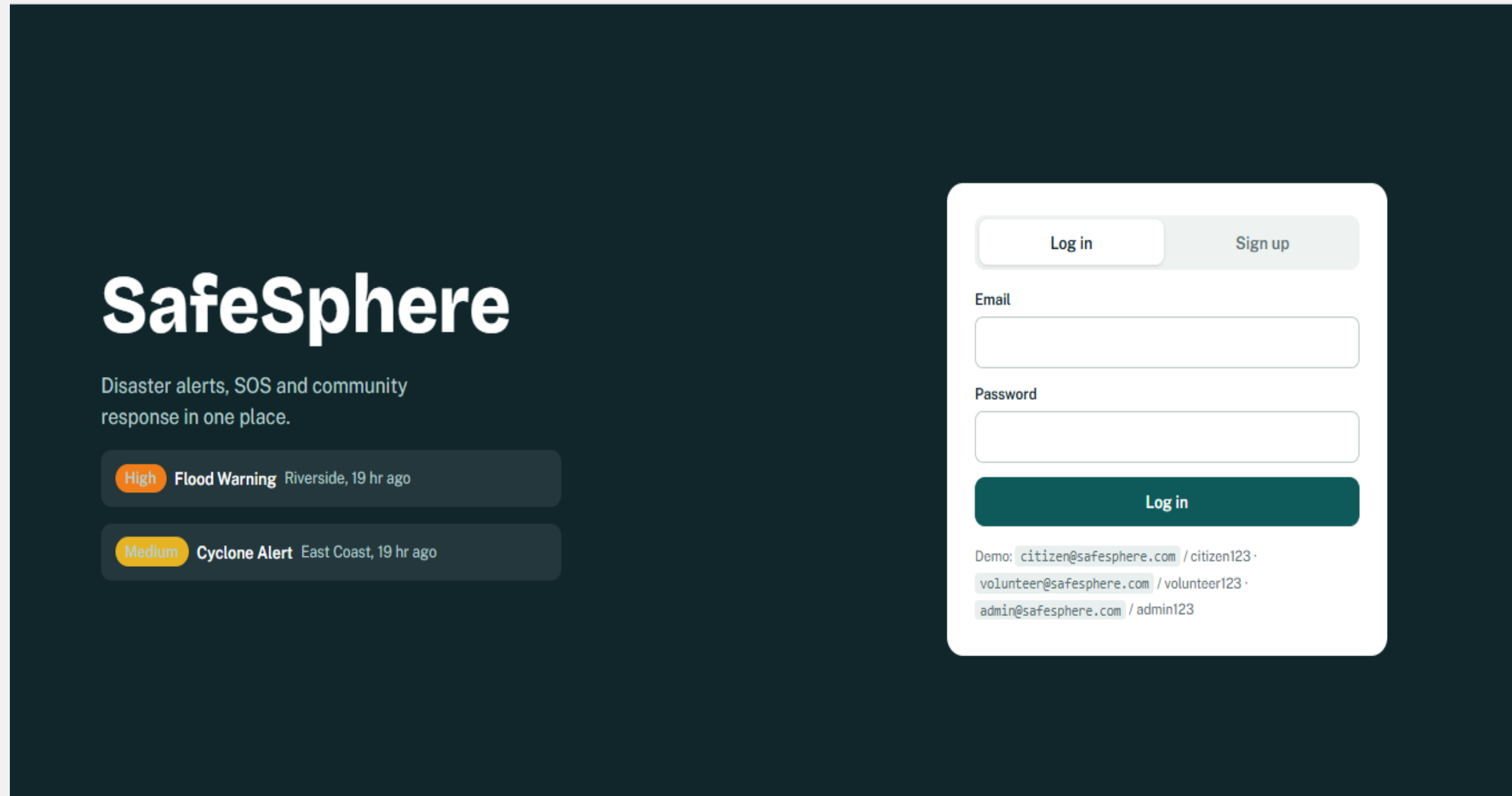


Frontend Design



- Home Page – introduces the platform and major emergency actions.
- Login / Registration – provides user authentication.
- Forms – used for incident reports and help requests.
- Dashboard – displays alerts, reports, assistance and updates.
- Navigation should make emergency actions easy to access.

Frontend Implementation



Dashboard

Alerts

SOS

Live map

Help requests

Family safety

Shelters & services

Live location on. Alerts are filtered to where you are.

SOS

In an emergency, press SOS

Shares your location, alerts your emergency contacts and the response team, and shows the nearest shelters and hospitals.

1

alerts near you

1

active incidents

120

slots at Community Hall (4.4 km away)

Safe

last check-in 18 hr ago

Earthquake Advisory

Earthquake in Central Zone, affects about 25 km

Minor tremors detected. Stay alert.

Safety: Drop, cover and hold on during shaking. Stay away from windows.

Evacuation: No evacuation needed.

20 hr ago · 5.7 km away

National Centre for Seismology · Verified

Citizen

Dashboard

Alerts

SOS

Live map

Help requests

Family safety

Shelters & services

Emergency SOS

Pick the emergency, then send. This alerts people right away, so use it for real emergencies.

Medical emergency

Trapped

Fire

Flood

Accident

Missing person

Other

Where are you (optional)

Street, landmark

Anything responders should know (optional)

Send SOS

SafeSphere

Citizen

Dashboard

Alerts

SOS

Live map

Help requests

Family safety

Shelters & services

Alerts

Only alerts that cover your current location are shown.

Earthquake Advisory

Earthquake in Central Zone, affects about 25 km

Minor tremors detected. Stay alert.

Safety: Drop, cover and hold on during shaking. Stay away from windows.

Evacuation: No evacuation needed.

20 hr ago · 5.7 km away

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Alerts

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Live map

Help requests

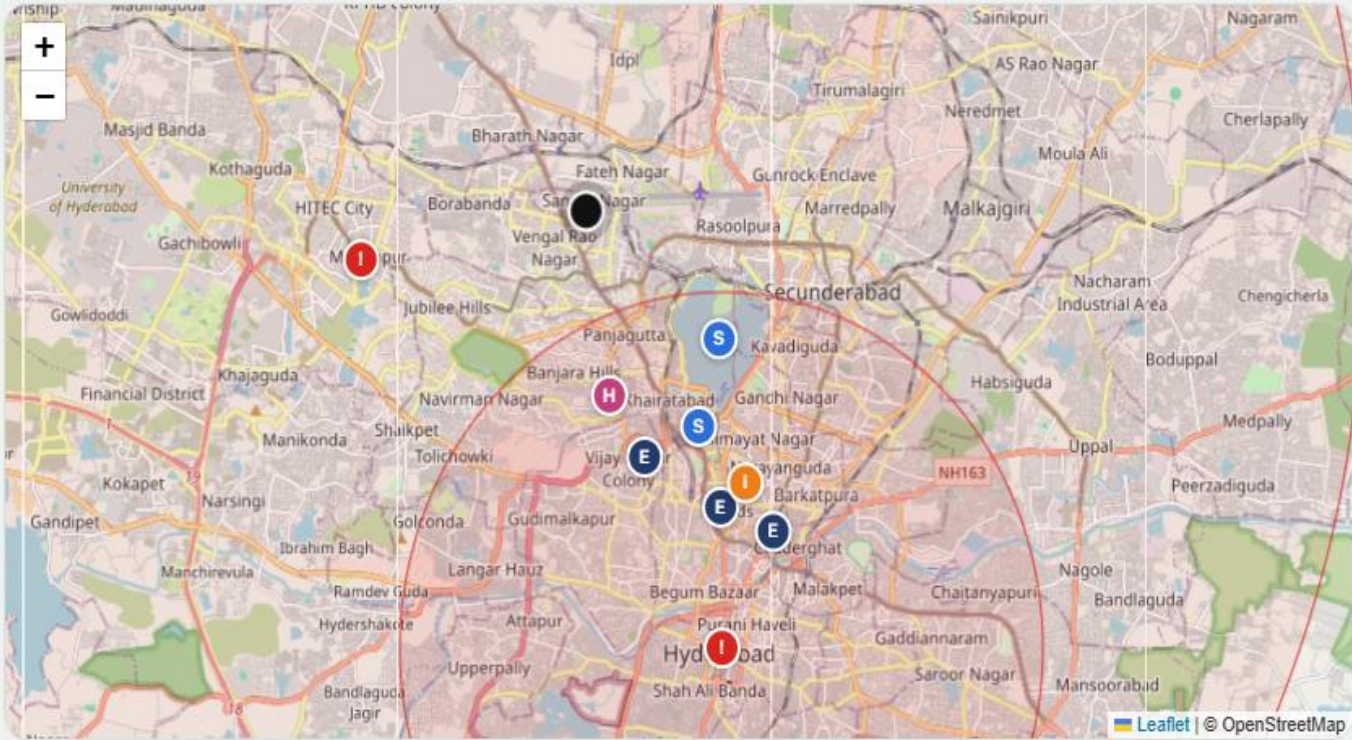
Family safety

Shelters & services

Live map

Tap a marker for details. Incidents show distance, severity and status.

Active disaster Reported incident Available volunteer Shelter Hospital Emergency service You



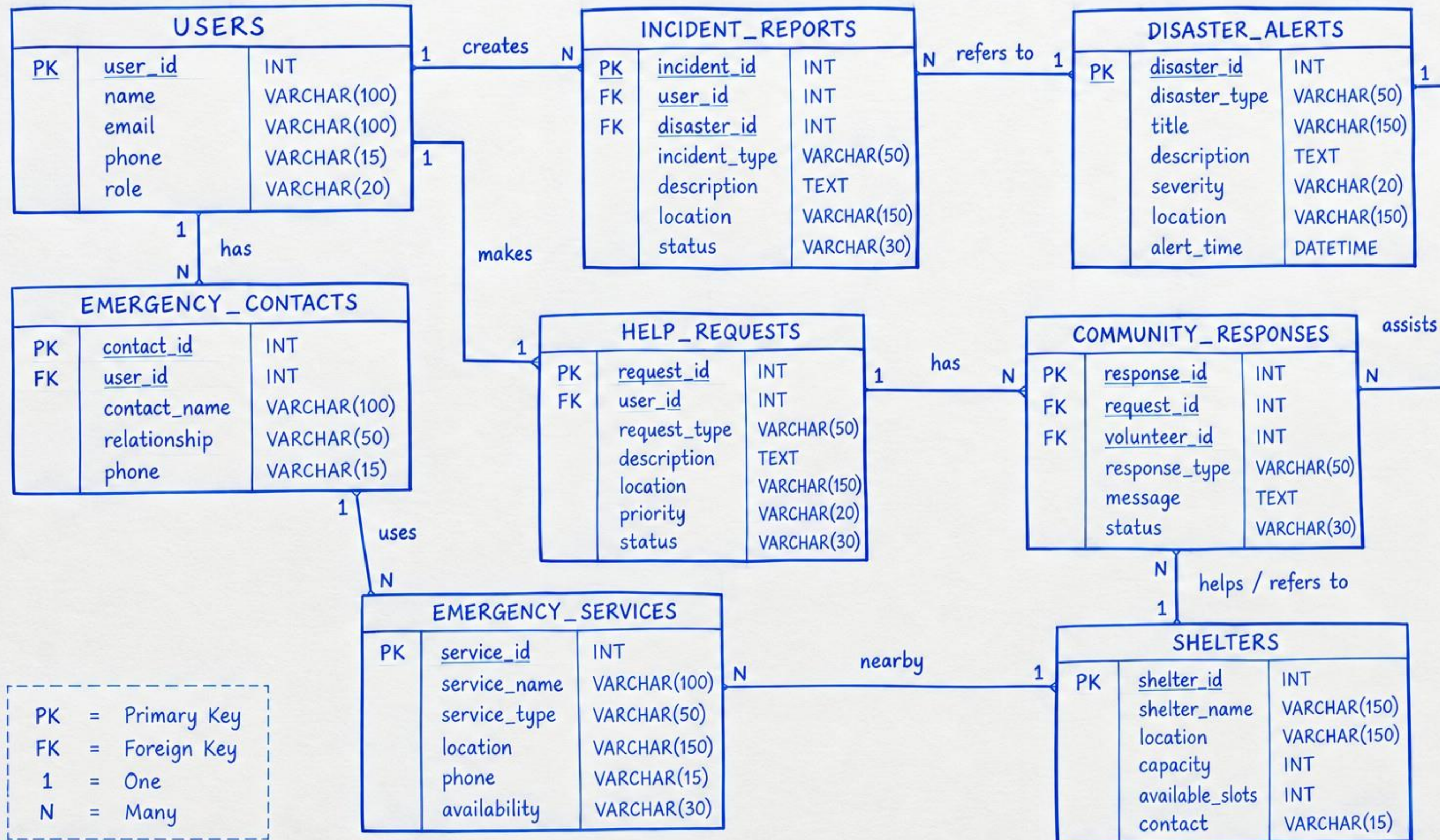
Database Design

- Users and emergency contacts
- Disaster alerts
- Incident reports
- Emergency/help requests
- Community or volunteer responses
- Shelter and emergency-service information



ER Diagram

ER Diagram - Disaster Alert & Community Response App



	contact_id	user_id	contact_name	relationship	phone
▶	1	1	Suresh Kumar	Father	9988776655
	2	2	Lakshmi Reddy	Mother	9988776656
	3	3	Meena Sharma	Mother	9988776657
	4	4	Raj Singh	Brother	9988776658
	5	5	Anil Rao	Father	9988776659
✱	NULL	NULL	NULL	NULL	NULL

	disaster_id	disaster_type	title	description	severity	location
▶	1	Flood	Heavy Flood Alert	Water levels are rising rapidly	High	Vijayawada
	2	Cyclone	Cyclone Warning	Strong winds expected in coastal areas	High	Visakhapatnam
	3	Fire	Industrial Fire	Fire reported near industrial area	Critical	Guntur
	4	Earthquake	Earthquake Alert	Earthquake reported in the region	Medium	Hyderabad
⌵	NULL	NULL	NULL	NULL	NULL	NULL

[illegible]

Current Progress

- Project title, abstract and key features defined.
- Frontend structure planned for home, authentication, forms and dashboard.
- Backend technology identified: Node.js.
- Database technologies identified: MySQL and MongoDB.
- REST API / OpenAPI identified for backend communication.
- Modules and architecture identified.
- Add screenshots/evidence for modules already implemented.

Challenges / Issues Faced

- Designing a simple interface for emergency situations.
- Managing location-based incident information.
- Keeping alerts and reports organized.
- Connecting frontend forms with backend services.
- Designing suitable database relationships.
- Ensuring authentication and emergency data are handled securely.

Future Work

- Implement and integrate all planned modules.
- Complete REST APIs and backend services.
- Add real-time alert delivery.
- Improve map-based disaster tracking.
- Integrate nearby shelters and emergency services.
- Strengthen authentication and authorization.
- Containerize and deploy using Docker.

Conclusion

- The proposed app provides a centralized platform for disaster communication.
- It combines alerts, incident reporting, help requests and community assistance.
- The planned architecture supports scalable and maintainable development.
- The project aims to improve reporting speed, coordination and community response.
- Next step: complete integration and demonstrate the working modules.